

1. Introduction

The objective of this project is to predict fair market prices of used cars using machine learning techniques.

Used-car prices depend on many different factors, so it is often difficult for buyers to determine whether a listed price is fair.

In this project, we built a regression-based machine learning system that predicts used-car prices and analyzes the price gap between predicted prices and actual listing prices.

This project includes the full machine learning workflow, from preprocessing and feature engineering to model evaluation and interpretation.

2. Dataset description

We used the Craigslist Vehicles Dataset from Kaggle.

The original dataset contains 426,880 records and 26 features, including both numerical and categorical variables.

Important numerical features include:

- year
- odometer
- latitude
- longitude

Important categorical features include:

- manufacturer
- condition
- fuel
- transmission
- drive
- type
- paint_color

- state

The dataset also contains dirty data such as missing values, duplicate rows, unrealistic prices, and high-cardinality categorical variables.

3. Data Inspection

Before preprocessing, we inspected the dataset using statistical and visual analysis.

We analyzed:

- missing value ratios
- target price distribution
- numerical feature correlations
- categorical feature frequencies

This process helped us understand the dataset structure and identify dirty data problems.

4. Data preprocessing

Missing value handling

Numerical missing values were replaced using median imputation, while categorical missing values were replaced using the most frequent category.

Duplicate removal

Duplicate rows were removed to reduce redundant information.

Invalid price removal

Listings with price values less than or equal to zero were removed because they are unrealistic for regression analysis.

Outlier Removal

We used the IQR (Interquartile Range) method to remove extreme outliers from features such as price, year, and odometer.

Rare category Grouping

To reduce dimensional explosion after one-hot encoding, rare categories were grouped into

"Other".

5. Feature Engineering

We created additional meaningful features to improve model performance.

The engineered features include:

- car_age:

Difference between the current year and vehicle year.

- odometer_per_year:

Mileage divided by vehicle age.

- is_high_mileage:

Binary feature indicating whether mileage exceeds 150,000.

These features were designed to better represent vehicle depreciation and usage patterns.

6. Scaling and Encoding

For numerical features, we compared three scaling methods:

- RobustScaler

- StandardScaler

- MinMaxScaler

Each scaler was evaluated using 3-fold cross-validation.

Categorical variables were encoded using One-Hot Encoding.

7. Model

We used RandomForestRegressor as the main learning algorithm.

Random Forest is an ensemble learning method that combines multiple decision trees to improve generalization performance and reduce overfitting.

To optimize the model, RandomizedSearchCV was used for hyperparameter tuning.

8. Evaluation Metrics

The model was evaluated using the following regression metrics:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- R^2 Score
- MAPE (Mean Absolute Percentage Error)

These metrics measure prediction accuracy from different perspectives.

9. Experimental Results

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Cleaned Dataset Shape: (358805, 20)
Sampled Dataset Shape: (10000, 20)

Numerical Features (7):
['year', 'odometer', 'lat', 'long', 'car_age', 'odometer_per_year', 'is_high_mileage']

Categorical Features (12):
['region', 'manufacturer', 'condition', 'cylinders', 'fuel', 'title_status', 'transmission', 'drive', 'size', 'type', 'paint_color', 'state']

=== Scaler Comparison (3-fold CV  $R^2$ ) ===
RobustScaler   : CV  $R^2$  = 0.6778
StandardScaler : CV  $R^2$  = 0.6778
MinMaxScaler   : CV  $R^2$  = 0.6777

Best Scaler: RobustScaler

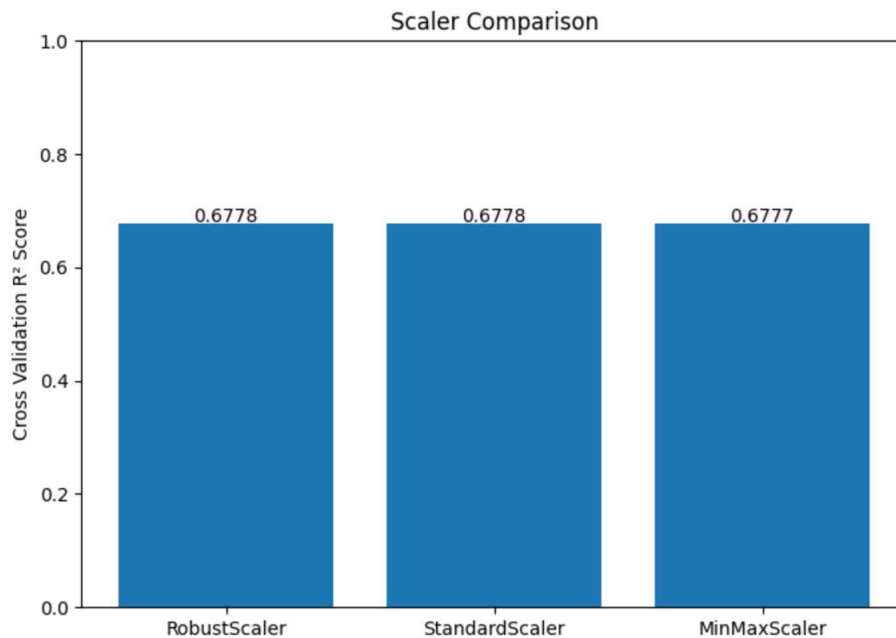
Best CV  $R^2$ : 0.6980

Best Parameters:
{'regressor__n_estimators': 50, 'regressor__min_samples_split': 2, 'regressor__min_samples_leaf': 1, 'regressor__max_depth': 15}

===== Model Evaluation =====
MAE   : 4557.13
RMSE  : 6965.00
 $R^2$   : 0.7057
MAPE  : 54.0800

===== Price Gap Analysis =====
      Actual_Price  Predicted_Price  Price_Gap  Deal_Label
61559          13710      20418.165835  -6708.165835  Overpriced
116925         49999      34181.629258  15817.370742  Underpriced
332746           799       7557.501460  -6758.501460  Overpriced
37693           8950      10648.106327  -1698.106327  Overpriced
331846         10995      12775.951361  -1780.951361  Overpriced
140781          2300       4604.287288  -2304.287288  Overpriced
42202          39990      15067.788635  24922.211365  Underpriced
89170          37590      23955.969830  13634.030170  Underpriced
50151          13990      14012.996047   -22.996047  Overpriced
356808         35590      28510.147395  7079.852605  Underpriced
PS C:\Users\pc\AppData\Local\Programs\Microsoft VS Code>
```

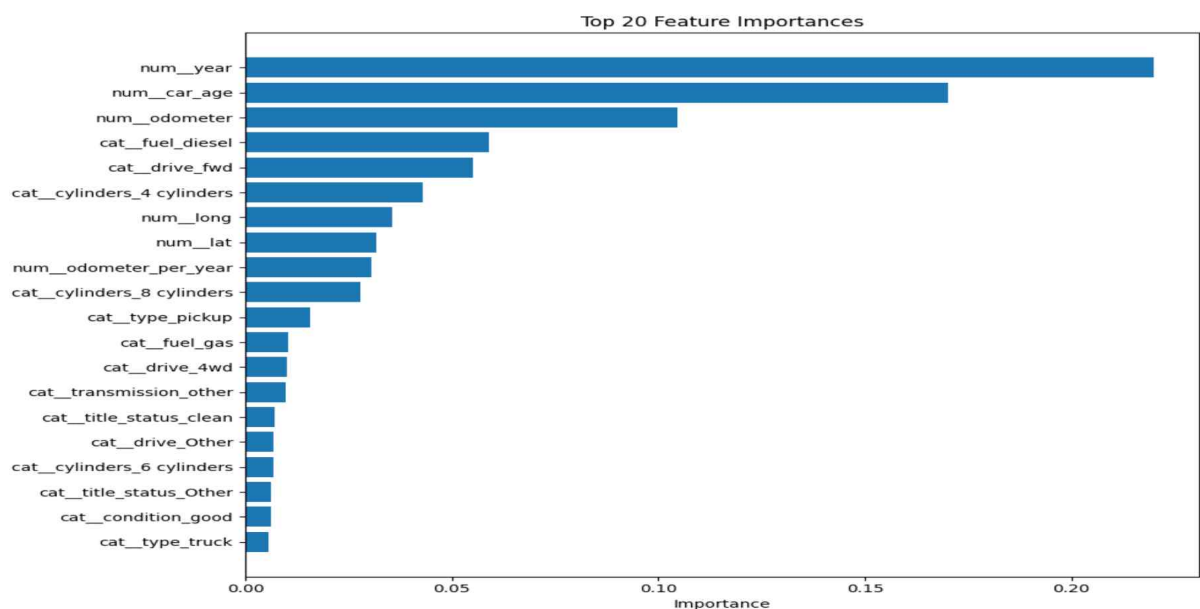
10. Scaler Comparison Analysis



The scaler comparison showed very similar performance across all scaling methods.

This result is expected because RandomForest is a tree-based model and is generally less sensitive to feature scaling than distance-based algorithms.

11. Feature Importance Analysis

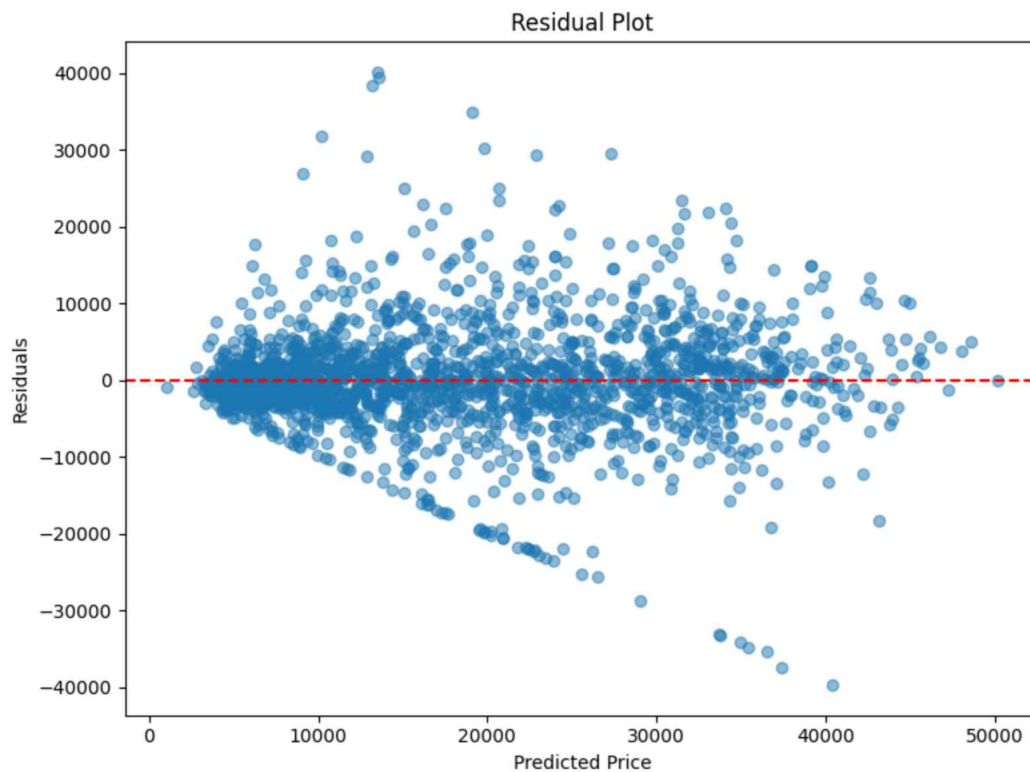


Feature importance analysis showed that vehicle age, mileage-related variables, and manufacturer-related categories strongly influenced vehicle price prediction.

This result is reasonable because vehicle age and mileage are strongly related to vehicle depreciation.

In particular, engineered features such as `car_age` and `odometer_per_year` contributed significantly to the model.

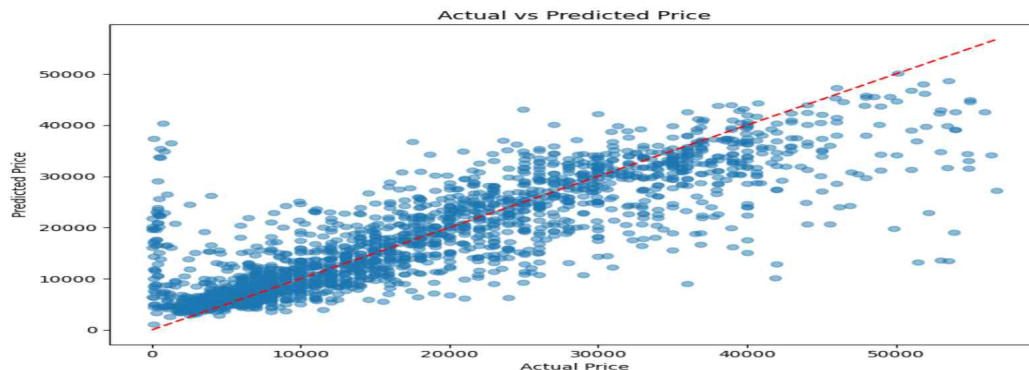
12. Residual Analysis



Residual analysis showed that most residuals were distributed around zero without strong systematic patterns.

Overall, the residual plot suggests that the model learned the general price trend reasonably well.

13. Price Gap Analysis



We analyzed the difference between predicted prices and actual listing prices.

If the predicted price was higher than the listed price, the vehicle was considered potentially underpriced.

If the predicted price was lower than the listed price, the vehicle was considered potentially overpriced.

This analysis can help users identify good or suspicious deals.

14. Limitations

This project has several limitations.

First, the dataset contains listed prices rather than actual transaction prices.

Second, important information such as accident history, maintenance history, and local demand was not available.

Third, Craigslist listings may contain noisy or inconsistent information.

15. Conclusion

In this project, we implemented an end-to-end machine learning pipeline for used-car price prediction.

The project included data cleaning, preprocessing, feature engineering, scaling comparison, model training, hyperparameter tuning, evaluation, and interpretation.

The final RandomForest model achieved reasonable prediction performance and provided meaningful price-gap analysis for used-car listings.